THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–10) | Class 6 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 17	The common difference is 3, so add 3 to the last term.
2	**(a)** acute	Compare 30° with 90° and 180°.
3	**(a)** 8531	Write the digits in decreasing order.
4	**(c)** 30	Multiply the symbols by the value of one symbol.
5	**(b)** 13	13 has only two factors, 1 and 13.
6	**(d)** 38	Perimeter $= 2(l + b)$.
7	**(b)** 8	The number below the line.
8	**(c)** 8	Diameter $= 2\times$ radius.
9	**(b)** 4	A regular polygon with 4 sides has 4 lines of symmetry.
10	**(c)** -7	The opposite has the same distance from 0 on the other side.
11	**(c)** 15	The 8th odd number is $2 \times 8 - 1$.
12	**(c)** 60	The hour hand turns 30° in 1 hour.
13	**(a)** 1358	Write the digits in increasing order (a non-zero digit first).
14	**(b)** 35	Multiply the height by the scale.
15	**(d)** 27	27 has more than two factors.
16	**(d)** 24	Perimeter $= 4\times$ side.
17	**(a)** $\frac{5}{7}$	Add the numerators; keep the denominator.
18	**(d)** 5	Radius $=$ diameter $\div 2$.
19	**(a)** 1	Letter A.
20	**(a)** -4	Add 1.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Common difference $= 5$ (1); Next three: $24, 29, 34$ (1).	1+1
22	It is acute (1); Compare with the right angle 90° (and the straight angle 180°) (1).	1+1
23	Largest 9752 (1); Smallest 2579 (1).	1+1
24	Each symbol is 10 students (1); $6 \times 10 = 60$ students (1).	1+1
25	Find pairs whose product is 16 (1); Factors: 1, 2, 4, 8, 16 (1).	1+1
26	Perimeter $= 2(l + b)$ (1); $2(17 + 9) = 52$ m (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Common difference $= 2$ (1); Term 8 $=$ first term $+7\times$ difference $= 3 + 7 \times 2$ (1); 17 (1).	1+1+1
28	$quarter$ of 360° (1); $= 90^\circ$ (1); Type: right (1).	1+1+1
29	Largest $= 741$, smallest $= 147$ (1); Sum $= 888$ (1); Difference $= 594$ (1).	1+1+1
30	Add all the numbers (1); Total $= 15 + 35 + 10 + 15 = 75$ (1); Hence the total is 75 (1).	1+1+1
31	Divide by primes in turn (1); $36 = 2 \times 2 \times 3 \times 3$ (1); Only prime factors appear (1).	1+1+1
	OR: Perimeter $= 4a = 4 \times 22 = 88$ m (1); Area $= a^2 = 22^2 = 484$ m^2 (1); State units (1).	1+1+1
32	Multiply numerator and denominator by 2, 3, 4 (1); $2/6$, $3/9$, $4/12$ (2).	1+2
	OR: Radius $= 5$ cm (1); Chord 7 cm vs diameter 10 cm (1); No: the diameter is the longest chord (1).	1+1+1

SECTION — D (Long Answer) — Answers — 6 × 5 = 30

Q.	Expected Answer / Value Points	Marks

33	First 6 terms: $2, 5, 8, 11, 14, 17$ (1); Term 7 is 20 and term 8 is 23 (2); Yes, 23 is the 8th term (2).	1+2+2
34	30° is acute (1); $360 \div 30 = 12$ (1); 12 turns make a full turn (1); Check: product is 360° (2).	1+1+1+2
35	Total $= 1245 + 1378 = 2623$ (1); Difference $= 1378 - 1245 = 133$ (2); Total strength is odd (2).	1+2+2
36	A: $4 \times 10 = 40$ (1); B: $6 \times 10 = 60$ (1); C: $5 \times 10 = 50$ (1); Total $= 150$ (2).	1+1+1+2
	OR: $12 = 2 \times 2 \times 3$ (1); $18 = 2 \times 3 \times 3$ (2); Common prime factors: 2, 3 (2).	1+2+2
37	Area of sheet $= 30 \times 20 = 600$ cm^2 (1); Area of square $= 8^2 = 64$ cm^2 (1); Area left $= 600 - 64 = 536$ cm^2 (1); Units: cm^2 (1); State the answer (1).	1+1+1+1+1
	OR: Used $=\frac{2}{5}\times 20 = 8$ m (1); Left $= 20 - 8 = 12$ m (1); Used + left = total (1); State the units (2).	1+1+1+2
38	Draw AB = 6 cm (1); Draw perpendicular at B and mark BC = 4 cm (1); Draw perpendicular at A and mark AD = 4 cm, join D and C (1); Check that DC = 6 cm (1); Perimeter $= 2(6 + 4) = 20$ cm (1).	1+1+1+1+1
	OR: Lines of symmetry $= 3$ (1); Order $= 3$ (1); Angle $= 360 \div 3 = 120^\circ$ (1); The polygon is a 3-sided regular polygon (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–10) | Class 6 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 14	The 7th even number is 2×7.
2	**(d)** 90	A full turn is 360°.
3	**(d)** 454	A palindrome reads the same forwards and backwards.
4	**(c)** 1 bundle of five and 2 single strokes	Every fifth stroke crosses the four before it, making a bundle of five.
5	**(b)** 6	Factors: 1, 2, 3, 4, 6, 12.
6	**(d)** 54	Area $= l \times b$.
7	**(c)** $\frac{5}{11}$	Subtract the numerators; keep the denominator.
8	**(a)** a compass	Property
9	**(c)** F	It cannot be folded into two matching halves.
10	**(d)** -4	Subtract 1.
11	**(b)** 36	$6 \times 6 = 36$.
12	**(c)** 3	Count every pair of points.
13	**(d)** 1000	Add 1.
14	**(c)** Grapes	Look for the largest number.
15	**(c)** 42	7×6
16	**(d)** 49	Area $= a \times a$.
17	**(c)** 8	Multiply the numerator by 4 too.
18	**(a)** 9	Opposite sides of a rectangle are equal.
19	**(a)** 180	Angle $= 360^\circ \div$ order.
20	**(a)** 5	Use the number line or tokens.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Denominators are equal (1); $\frac{5}{9}$ (1).	1+1
22	Diameter $= 2 \times 6 = 12$ cm (1); Longest chord is the diameter, 12 cm (1).	1+1
23	A regular polygon has as many lines as sides (1); 4 lines (1).	1+1
24	Opposite of 5 is -5 (1); Opposite of -5 is 5 (1).	1+1
25	Squares are $1 \times 1, 2 \times 2, \ldots$ (1); $1, 4, 9, 16, 25$ (1).	1+1
26	It turns 30° per hour (1); $3 \times 30 = 90^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Lines $= 4$ (1); Order $= 4$ (1); Smallest angle $= 90^\circ$ (1).	1+1+1
28	Subtracting an integer is the same as adding its opposite (1); $5 + (-8)$ (1); $= -3$ (1).	1+1+1
29	$1 + 3 + 5 + 7 + 9 = 25$ (1); $25 = 5 \times 5$ is a square number (2).	1+2
30	Each hour mark is 30° apart (1); Angle $= 1 \times 30 = 30^\circ$ (1); Type: acute (1).	1+1+1
31	Reverse $= 21$ (1); Sum $= 12 + 21 = 33$ (1); Yes, palindrome (1).	1+1+1
	OR: Count each value (1); 1: 2; 2: 5; 3: 4; 4: 4 (1); Total $= 15$ (1).	1+1+1
32	By 2: no (last digit 5) (1); By 3: yes (digit sum 9) (1); By 5: yes (1).	1+1+1
	OR: Perimeter $= 2(60 + 40) = 200$ m (1); Wire needed 200 m (1); Area $= 60 \times 40 = 2400$ m^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 6 × 5 = 30

Q.	Expected Answer / Value Points	Marks

33	After going down: $2 - 5 = -3$ (1); After going up: $-3 + 1 = -2$ (1); Final floor -2 (1); Below ground (2).	1+1+1+2
34	$1 + 3 + 5 + 7 = 16$ (1); $16 = 4 \times 4$ (2); Sum of first 5 odd numbers $= 5 \times 5 = 25$ (2).	1+2+2
35	Segments: $4 \times 3 \div 2 = 6$ (1); A can be paired with each of the other 3 points (1); 3 segments have A as an end point (1); A segment has two end points and a fixed length (2).	1+1+1+2
36	Greatest $= 8642$ (1); Smallest $= 2468$ (1); Difference $= 6174$ (1); Even (2).	1+1+1+2
	OR: Monday 30 (1); Tuesday 50 (1); Wednesday 40 (1); Total 120 books (2).	1+1+1+2
37	Factors: 1, 2, 4, 7, 14, 28 (1); Count $= 6$ (1); Sum of all factors $= 56$ (1); Even (2).	1+1+1+2
	OR: Side $= 10$ m (1); Rectangle breadth: $2(12 + b) = 40$ gives $b = 8$ m (1); Square area $= 100$ m^2 (1); Rectangle area $= 96$ m^2 (1); The square has the larger area (1).	1+1+1+1+1
38	Add the numerators: $3 + 4 + 5 = 12$ (1); Sum $= \frac{12}{7}$ (1); As a mixed number: $1\frac{5}{7}$ (1); Reduce if possible (2).	1+1+1+2
	OR: Draw AB = 4 cm (1); Raise perpendiculars at A and B of length 4 cm (1); Join the ends to complete the square (1); Perimeter $= 16$ cm (1); Sum of angles $= 4 \times 90^\circ = 360^\circ$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–10) | Class 6 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** 10	$\frac{4\times5}{2} = 10$.
2	**(b)** two end points	Definition
3	**(c)** 999	Subtract 1.
4	**(b)** Football	Look for the smallest number.
5	**(d)** 363	Divisibility test of 3.
6	**(a)** 15	All three sides are equal.
7	**(c)** 36	Divide 48 by 4 and multiply by 3.
8	**(d)** 90	All angles of a square are right angles.
9	**(c)** infinitely many lines of symmetry	Symmetry property
10	**(d)** 2	The number to the right on the number line is greater.
11	**(c)** 8	$2 \times 2 \times 2 = 8$.
12	**(b)** 30	In 1 minute the minute hand turns 6°.
13	**(b)** 400	Place value = digit × value of its place.
14	**(b)** 2	Count (tally) the number of times.
15	**(c)** 10	Add the primes 2, 3, 5, 7, ... (1 is not prime).
16	**(d)** 24	Six equal sides.
17	**(d)** $\frac{7}{3}$	2×3+1 = 7, so the fraction is 7 over 3.
18	**(a)** 360	4 right angles.
19	**(d)** 3	Order = number of times it fits.
20	**(b)** 6	Distance is never negative.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Reverse of 636 is 636 (1); Yes, it is a palindrome (1).	1+1
22	Draw strokes; every fifth stroke crosses four (1); 9 = 1 bundle of five and 4 single strokes (1).	1+1
23	Multiply 6 by 1, 2, 3, 4, 5 (1); 6, 12, 18, 24, 30 (1).	1+1
24	Area $= l \times b$ (1); $14 \times 9 = 126$ cm^2 (1).	1+1
25	$17 \div 5$ gives quotient 3, remainder 2 (1); $3\frac{2}{5}$ (1).	1+1
26	A rectangle (1); Other sides are 8 cm and 5 cm (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Equal denominators: compare numerators (1); $\frac{5}{11} < \frac{8}{11}$ (1); Greater: $\frac{8}{11}$ (1).	1+1+1
28	Open the compass to 3 cm using the ruler (1); Fix the pointer at the centre O (1); Turn the pencil to complete the circle; every point is 3 cm from O (1).	1+1+1
29	Imagine folding the letter (1); A: yes; H: yes; B: yes; C: yes (1); Check each fold (1).	1+1+1
30	Negative numbers are smaller than 0 and 0 is smaller than positive numbers (1); Ascending: -7, -2, 0, 3, 5 (1); Further from 0 on the negative side means smaller (1).	1+1+1
31	$1 + 2 + \cdots + 9$ (1); $= 45$ (2).	1+2
	OR: It turns 6° per minute (1); $8 \times 6 = 48^\circ$ (1); Less than 90° (1).	1+1+1
32	Middle row: $3 + 5 = 8$ and $5 + 2 = 7$ (1); Top number $= 8 + 7$ (1); $= 15$ (1).	1+1+1
	OR: Most: Tennis (50) (1); Least: Football (25) (1); Difference $= 25$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Check both folds (1); (a) H, O, X (1); (b) H, O, X, Z, N (1); Order 2 means it fits after a half turn (2).	1+1+1+2
34	Descending: 6, 3, -1, -4 (1); Smallest -4, largest 6 (1); Sum $= 2$ (1); Mark each at its distance from 0 (2).	1+1+1+2
35	First 6 terms: $4, 9, 14, 19, 24, 29$ (1); Term 7 is 34 and term 8 is 39 (2); Yes, 39 is the 8th term (2).	1+2+2
36	45° is acute (1); $360 \div 45 = 8$ (1); 8 turns make a full turn (1); Check: product is 360° (2).	1+1+1+2
	OR: Total $= 2310 + 1795 = 4105$ (1); Difference $= 2310 - 1795 = 515$ (2); Total strength is odd (2).	1+2+2
37	A: $7 \times 5 = 35$ (1); B: $9 \times 5 = 45$ (1); C: $6 \times 5 = 30$ (1); Total $= 110$ (2).	1+1+1+2
	OR: $20 = 2 \times 2 \times 5$ (1); $30 = 2 \times 3 \times 5$ (2); Common prime factors: 2, 5 (2).	1+2+2
38	Area of sheet $= 40 \times 25 = 1000$ cm^2 (1); Area of square $= 10^2 = 100$ cm^2 (1); Area left $= 1000 - 100 = 900$ cm^2 (1); Units: cm^2 (1); State the answer (1).	1+1+1+1+1
	OR: Used $=\frac{3}{8}\times 40 = 15$ m (1); Left $= 40 - 15 = 25$ m (1); Used + left = total (1); State the units (2).	1+1+1+2

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